

LINCOSURE 400MG/G POWDER

Description and Composition

LINCOSURE 400MG/G POWDER is a yellowish-white powder. Each gram contains 400mg of Lincomycin (as hydrochloride) as its active ingredient.

Pharmacodynamics

Lincomycin is a lincosamide antibiotic derived from *Streptomyces lincolensis* which inhibits protein synthesis. Lincomycin binds to the 50S sub-unit of the bacterial ribosome close to the peptidyl transfer centre and interferes with the peptide chain elongation process by causing premature peptidyl-tRNA dissociation from the ribosome. Lincomycin is active against some Gram-positive bacteria (*Clostridium perfringens*) and mycoplasmas (*Mycoplasma hyopneumoniae*).

While the lincosamides are generally considered to be bacteriostatic agents, the activity depends on the sensitivity of the organism and concentration of the antibiotic. Lincomycin may be either bactericidal or bacteriostatic. Resistance to lincomycin is frequently conferred by plasmid-borne factors (*erm* genes) coding for methylases modifying the ribosomal binding site and frequently leading to cross-resistance to other antimicrobials of the macrolides, lincosamides and streptogramins group. However, the most prevalent mechanism in mycoplasmas is the alteration of the binding site through mutational events (chromosomal resistance). Lincomycin resistance mediated by efflux pumps, or by inactivating enzymes, has also been described. There is often complete cross-resistance between lincomycin and clindamycin.

Pharmacokinetics

In pigs, lincomycin is rapidly absorbed following oral administration. A single oral administration of lincomycin hydrochloride, at dose levels of approximately 22, 55 and 100mg per kg body weight in pigs, resulted in dose related lincomycin serum levels, detected for 24 to 36 hours after administration. Peak serum levels were observed at 4 hours after dosing. Similar results were observed following single oral doses of 4.4 and 11.0mg per kg body weight in pigs. Levels were detectable for 12 to 16 hours, with peak concentrations occurring at 4 hours. A single oral dose of 10mg per kg body weight was administered to pigs to determine the bioavailability. The oral absorption of lincomycin was found to be 53% $\pm$ 19%. Repeated dosing of pigs with daily oral doses of 22mg lincomycin per kg body weight for 3 days indicated no accumulation of lincomycin in the species, with no detectable serum levels of antibiotic after 24 hours post administration.

Crossing the intestinal barrier, lincomycin is widely distributed to all tissues, especially the lungs and joint cavities; the volume of distribution is about 1 litre. The elimination half-life of lincomycin is greater than 3 hours. Approximately 50% of lincomycin is metabolised in the liver. Lincomycin undergoes enterohepatic circulation. Lincomycin is eliminated unchanged or in the form of various metabolites in bile and urine. High concentrations of the active form are observed in the intestine.

Chickens were administered lincomycin hydrochloride in the drinking water at a level of approximately 34mg per litre (5.1 to 6.6 mg per kg body weight) for seven days. Metabolites comprised more than 75% of total residues in the liver. Unmetabolised lincomycin declined at a slightly faster half-life ($t_{1/2}$ = 5.8 hours) than total residue. Lincomycin and one unknown metabolite comprised >50% of the muscle residue at zero hours. The excreta contained mostly unmetabolised lincomycin (60 to 85%) during treatment.

Indication

Pigs:

Treatment and metaphylaxis of enzootic pneumonia caused by *Mycoplasma hyopneumoniae*.
The presence of the disease in the group must be established before the product is used.

Chickens:

Treatment and metaphylaxis of necrotic enteritis caused by *Clostridium perfringens*.
The presence of the disease in the group must be established before the product is used.

Recommended Dose

For use in drinking water. Dosing guidance and recommended doses:

To ensure a correct dosage, body weight should be determined as accurately as possible to avoid underdosing. The intake of medicated water depends on the physiological and clinical condition of the animals. In order to obtain the correct dosage, the concentration of the lincomycin has to be adjusted accordingly. The uptake of water should be monitored frequently. The medicated water should be the only source of drinking water for the animals for the entire duration of the treatment period. After the end of the medication period, the water supply system should be cleaned appropriately to avoid intake of sub-therapeutic amounts of the active substance.

Pigs:

Enzootic pneumonia: 10mg lincomycin per kg of body weight (corresponding to 25mg product per kg body weight) for 21 consecutive days.

Chickens:

Necrotic enteritis: 5mg lincomycin per kg of body weight (corresponding to 12.5mg product per kg body weight) for 7 consecutive days.

The concentration to be used depends on the actual body weight and the water consumption of the animals and can be calculated according to the following formula:

$$\frac{\text{Dosage (mg product per kg weight per day)}}{\text{Average daily water intake (litre per animal)}} \times \frac{\text{Mean body weight (kg) of animals to be treated}}{\text{Average daily water intake (litre per animal)}} = \dots \text{ mg product per litre drinking water}$$

Average daily water intake (litre per animal)

The use of suitably calibrated weighing equipment is recommended if part packs are used. The daily amount is to be added to the drinking water in such a way that all medication will be consumed within 24 hours. Medicated drinking water should be freshly prepared every 24 hours. No other source of drinking water should be available.

Route of Administration

For oral administration via drinking water.

Contraindication

Do not use in cases of known hypersensitivity to the active substance or to any of the excipients.

Do not use in cases of known resistance to lincosamides.

Do not administer, and do not allow access to water containing lincomycin to horses, ruminants, guinea pigs, hamsters, chinchillas and rabbits as this could result in severe gastro-intestinal disturbance.

Do not use in cases of hepatic dysfunction.

Warning and Precautions

Special warnings for each target species:

Medicated drinking water uptake can be affected by the severity of the disease. In case of insufficient uptake of water, pigs should be treated parenterally.

The susceptibility of *Mycoplasma hyopneumoniae* to antimicrobial agents is difficult to test *in vitro* owing to technical constraints. In addition, there is a lack of clinical breakpoints for both *M. hyopneumoniae* and *C. perfringens*. Where possible, therapy should be based on local (regional, farm level) epidemiological information concerning the response of enzootic pneumonia/necrotic enteritis to treatment with lincomycin.

Special precautions for use:

Special precautions for use in animals:

Use of the veterinary medicinal product preferably should be based on identification of the target pathogen and susceptibility testing of the bacteria isolated from the animal. However, also see text under

section 'Special warnings for each target species'. Official, national and regional antimicrobial policies should be taken into account when the veterinary medicinal product is used.

Use of the veterinary medicinal product deviating from the instructions given may increase the prevalence of bacteria resistant to the lincomycin and may decrease the effectiveness of treatment with other lincosamides, macrolides and streptogramin B due to the potential for cross-resistance.

Repeated or prolonged use should be avoided by improving the farm management and hygiene practices.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

This product contains lincomycin and lactose, either of which can cause allergic reactions in some people. People with known hypersensitivity to lincomycin or any other lincosamide, or to lactose, should avoid contact with the veterinary medicinal product.

Care should be taken not to raise and inhale any dust.

Contact with skin and eyes should be avoided.

Personal protective equipment consisting of approved dust masks (either a disposable half-mask respirator or a non-disposable respirator), gloves and safety glasses should be worn when handling and mixing the product. If respiratory symptoms develop following exposure, seek medical advice and show this warning to the physician.

In case of accidental exposure to the skin, eyes or mucous membranes, wash the affected area thoroughly with plenty of water. If symptoms such as skin rash or persistent eye irritation appear after exposure, seek medical advice immediately and show the package leaflet or label to the physician.

Do not smoke, eat or drink when handling the product.

Wash hands and any exposed skin with soap and water immediately after use.

Interaction with Other Medicaments

Antagonism may exist between lincomycin and macrolides such as erythromycin and other bactericidal antibiotics; concurrent use is therefore not recommended due to competitive binding at the 50S ribosomal subunit of the bacterial cell. The bioavailability of lincomycin may decrease in the presence of gastric antacids or activated charcoal, pectin or kaolin. Lincomycin can potentiate neuromuscular effects of anaesthetics and muscle relaxants.

Pregnancy and Lactation

Laboratory studies in rats have not produced any evidence of teratogenic effects, although foetotoxicity has been reported. The safety of the veterinary medicinal product has not been established during pregnancy, lactation or lay in the target species. Use only according to the benefit/risk assessment by the responsible veterinarian.

Side Effects

On rare occasions, pigs given lincomycin-medicated water may develop diarrhoea/ soft stools and/or mild swelling of the anus within the first two days after onset of treatment. On rare occasions some pigs may show reddening of the skin and mild irritable behaviour. These conditions are usually self-correcting within 5 to 8 days without discontinuing the lincomycin treatment. Allergic/hypersensitive reactions occur on rare occasions.

The frequency of adverse reactions is defined using the following convention:

- very common (more than 1 in 10 animals treated displaying adverse reactions)
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

Symptoms and Treatment of Overdose

A dosage greater than 10mg lincomycin per kg of body weight may cause diarrhoea and loose stools in pigs. In case of accidental overdose, the treatment must be stopped and restarted at the recommended dose level. There is no specific antidote, treatment is symptomatic.

Storage Condition

Store below 30°C.

Shelf life

3 years.

To be used within 24 hours after dilution in drinking water.

After first opening the container, any product remaining after 24 hours should be discarded.

Withdrawal Periods

Meat and offal:

Pigs: 1 day

Chickens: 5 days

Not authorised for use in laying birds producing eggs for human consumption.

Packing

100g, 200g, 500g, 1kg.

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